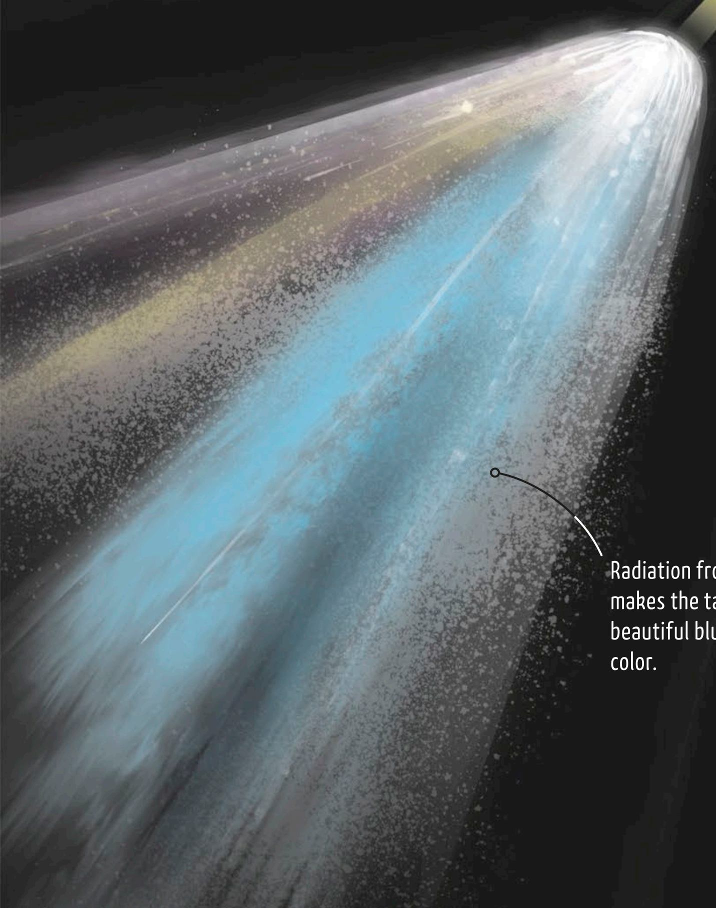
Sometimes comets slam into planets that are in their way. The comet explodes in the atmosphere or crashes into the planet's surface.



## Return

If it doesn't fizzle out or hit a planet, the comet will travel back to the Kuiper Belt, getting slower and slower until gravity pulls it back toward the sun in an elliptical (oval-shaped) orbit. It can take as many as 200 years for a comet to complete one rotation around the sun.

## Fizzling out

If the comet gets too close to the sun, it overheats, and the frozen gas and ice turn straight into gas. This process is called sublimation and causes the comet to fizzle out.

Radiation from the sun makes the tail glow a beautiful blue-white color.

*In ancient Greek, "comet" means "head with long hair."*

## Speeding comet

A comet knocked out of the Kuiper Belt will start to speed toward the sun. As it gets closer, it heats up, and the frozen gases and water turn back into gas, creating a small atmosphere called a coma. This coma gets pushed back by a stream of material from the sun called the solar wind, forming the tail of the comet.

*See also*
Find out more about the solar system (14–15).